Making Herding Measurable: A Game-Theoretic Framework Diagnoses and Treats LLM Review Panels

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Abstract. Multi-agent large language model (LLM) systems reach collective judgments through output aggregation, yet it remains poorly understood whether such panels exhibit the herding pathologies long documented in human committees—and if they do, which mechanisms can repair them. Answering these questions requires two instruments: a way to *measure* collective behavior and a way to *intervene* on it. We show that a previously developed game-theoretic consensus framework provides exactly these instruments: its AEIC (Assumption–Evidence–Inference–Conclusion) decomposition turns herding into a measurable quantity through an explicit Conclusion field, while its three-layer Stackelberg mechanism maps onto the three loci of intervention—aggregation-time calibration, discussion-time regulation, and per-instance protocol routing. We subject the framework to a *diagnose-then-intervene* validation on automated scholarly review, a domain where consensus quality is inherently behavioral. **Study 1** diagnoses 300 stratified ICLR 2024 papers reviewed by homogeneous five-reviewer LLM panels under three aggregation conditions. Independent LLM reviews exhibit a systematic reject bias (14.4% accept rate on clear-accept papers), and three rounds of unstructured discussion trigger a *negative ratchet*: minority positions flip toward rejection by 96:0 (binomial p=2.5×10^{-29}; 78% elimination), panel rating variance collapses by 51% (Wilcoxon p=3.9×10^{-32}), and mean ratings drift downward—while decision agreement with the official outcome never improves (61.7%→59.7%, n.s.). A cross-model replication (DeepSeek-V3, GLM-4-Flash, DeepSeek-R1) reveals *opposite pathological attractors*: V3 and R1 panels ratchet toward rejection whereas GLM panels ratchet toward acceptance with 94% hedged conclusions; the strongest reasoner (R1, 67.3% agreement) suffers the largest diversity collapse (76%). **Study 2** applies the framework's three layers as interventions, each targeting one diagnosed pathology. Calibrated Stackelberg aggregation (θ^*) learns a positive bias term ($\tau = +1.04$) that repairs the reject bias (clear-accept agreement 2%→38%), though simple re-thresholding captures most of the gain. Prompt-level behavioral regulation—evidence-anchored minority persistence—leaves the ratchet essentially intact (elimination 78%→71%) while preserving decision quality, an honest negative result indicating that rhetorical interventions cannot overcome capability-driven conformity. Per-paper protocol routing matches the best fixed protocol (64.7%) at 56% lower cost, demoting

discussion to an on-demand resource. Together, the studies establish a calibration-first doctrine for LLM consensus systems and demonstrate that behavioral health metrics—minority survival, variance preservation, drift—are far more sensitive endpoints than decision accuracy for panel-level evaluation.

Keywords: Multi-agent consensus · Stackelberg game · LLM peer review · Herding · Sycophancy · Calibrated aggregation

1 Introduction

Multi-agent systems powered by large language models (LLMs) increasingly perform *collective judgment*: multiple model instances independently analyze a problem, exchange positions, and aggregate into a consensus decision [13,29,31,9]. Yet evaluation of such systems remains one-dimensional: accuracy of the final decision. *How* the panel gets there—whether it preserves legitimate dissent, maintains rating diversity, and converges for the right reasons—is rarely measured, because free-text deliberation makes these behavioral quantities hard to extract.

This neglect is risky. Human committees are known to herd: discussion among reviewers is associated with score compression and conformity effects, and initial majority positions disproportionately determine outcomes. Whether LLM panels exhibit analogous pathologies—and with what direction and magnitude—has remained largely uncharacterized. The question matters for mechanism design: if herding exists, its structure dictates which class of intervention can repair it. Aggregation-time corrections suffice if the pathology is a static bias; behavioral regulation is required if discussion itself destroys information; and routing is appropriate if deliberation only pays off on a subset of instances.

Answering either question requires instruments that conventional pipelines lack: herding must be made *measurable* before it can be treated. In this paper we show that a previously developed game-theoretic consensus framework provides exactly these instruments. Its **AEIC decomposition** (Section 3) structures each agent's output into Assumption, Evidence, Inference, and an explicit Conclusion field—a stance probe that renders collective behavior directly measurable (+28pp extraction reliability over unstructured text). Its **three-layer Stackelberg incentive mechanism** (Section 4), with formal ϵ-incentive compatibility (Section 6), maps onto the three loci of intervention: static parameter optimization at aggregation time, online and dynamic regulation during discussion, and **adaptive protocol routing** per instance (Section 5). On QA benchmarks, the framework already answered the *decision-quality* question, achieving 87.6% consensus accuracy through its router; here we subject it to the *behavioral-health* question on automated scholarly review—a domain that is high-stakes and intrinsically behavioral, where LLM agents already generate review feedback on research papers [18,15].

The validation follows a *diagnose-then-intervene* design (Section 7). **Study 1** assembles 300 stratified ICLR 2024 papers (clear-accept / borderline / clear-reject) and homogeneous five-reviewer LLM panels under three conditions—

independent aggregation, three-round discussion, and meta-review synthesis. The diagnosis confirms three pathologies with strong statistical evidence: a systematic *reject bias* in independent reviews (14.4% accept rate on clear-accept papers); a *negative ratchet* during discussion, where minority positions flip toward rejection by 96:0 (p=2.5×10^{-29}) with 78% elimination and a 51% collapse of rating variance (p=3.9×10^{-32}); and a complete absence of decision benefit from discussion (61.7%→59.7% agreement, n.s.). A cross-model replication on DeepSeek-V3, GLM-4-Flash, and DeepSeek-R1 reveals that herding is universal but its *direction* is model-dependent: V3 and R1 panels ratchet toward rejection while GLM panels ratchet toward acceptance, and the strongest reasoner (R1) suffers the largest diversity collapse (76%)—capability does not protect against herding.

Study 2 then deploys the framework's three layers as targeted interventions, one per pathology. The static layer (*calibrated Stackelberg aggregation*, θ^*) learns a positive calibration term $\tau = +1.04$ and lifts clear-accept agreement from 2% to 38%, though a simple re-thresholded mean captures most of the gain—calibration is necessary and nearly sufficient. The dynamic layer (*evidence-anchored behavioral regulation*) leaves the negative ratchet essentially intact (elimination 78%→71%; intervened minorities persist at 29% vs. 28%) while preserving decision quality—an honest negative result showing that prompt-level regulation cannot overcome capability-driven conformity. The routing layer (*per-paper protocol selection*) matches the best fixed protocol (64.7%) at 56% lower inference cost, establishing deliberation as an on-demand resource rather than a default.

Our contributions are:

1. **A large-scale paired diagnosis of LLM review panels.** Three pathologies—reject bias, negative ratchet, diversity collapse—established with paired designs and strong statistical evidence, using behavioral health metrics (minority survival, variance preservation, drift) that prove far more sensitive than decision accuracy.
2. **Cross-model generality with opposite pathological attractors.** Herding direction is model-dependent (reject-ratchet vs. accept-sycophancy), and stronger reasoning capability co-occurs with stronger diversity collapse.
3. **A calibration-first intervention doctrine.** Aggregation-time calibration repairs decision bias cheaply; prompt-level behavioral regulation fails to restore diversity; per-paper routing preserves accuracy at 56% lower cost—together mapping which mechanism class treats which pathology.

2 Related Work

Multi-Agent LLM Consensus. The aggregation of multiple LLM outputs has been studied through several paradigms. Self-consistency [31] samples multiple reasoning paths and selects the most frequent answer. Multi-agent debate [9,6] extends this to iterative discussion among multiple model instances. ChatEval [3]

applies this to evaluation tasks. Zhao et al. [38] propose cardinal voting based on social choice theory. Li et al. [17] introduce dynamic weighting.

A critical failure mode in these systems is *sycophancy*—agents adopting majority positions without genuine reasoning. Bertalanič and Fortuna [2] provide the most comprehensive empirical characterization to date, documenting three failure pathways in homogeneous multi-agent debate (n=10): sycophantic conformity (up to 85.5% modal adoption), contextual fragility (up to 70% of previously correct agents destabilized by peer rationales), and consensus collapse (up to 32.3pp oracle gap where plurality voting discards correct answers). Their central finding—that homogeneous teams without structured roles do not benefit from unguided peer exchange—directly motivates our work on structured role enforcement through AEIC decomposition and behavioral regulation through Stackelberg mechanisms. Liu et al. [20] further show that expanded agent memory can *degrade* cooperation in game-theoretic settings, with longer recall reducing cooperative intent in 18 of 28 model–game configurations—a finding consistent with our result that quality signals cannot reliably distinguish correct from misleading knowledge even when agents have access to differentiated information.

Pitre et al. [23] address sycophancy through ConsensAgent, which dynamically refines prompts based on agent interactions to mitigate herding. ConsensAgent operates at the prompt level within a single debate topology. Our dynamic regulation differs fundamentally: rather than optimizing prompts for individual agents, we monitor *collective behavioral dynamics* (stance change rate, majority ratio, convergence speed) across rounds and apply differentiated interventions—parameter adjustment *and* prompt modification—targeted at the specific pathology detected (sycophantic vs. deadlock). Swamy et al. [28] provide formal metrics for sycophancy in multi-agent debate; their finding that sycophancy levels vary substantially across agent roles informs our stance-change-based detection. Most recently, Kumarappan et al. [16] demonstrate that multi-agent sycophancy is structural (present in base models, not just RLHF artifacts) and that mitigations should target "structured dissent at the pipeline level"—this is precisely what our Stackelberg dynamic regulation achieves through behavioral state detection and adaptive EV-weight suppression.

LLM-as-Reviewer and Panel Behavior. LLMs are increasingly used to generate scholarly review feedback. Liang et al. [18] show that GPT-4 provides useful paper feedback in a large user study, and AgentReview [15] simulates the full peer review process with LLM agents, finding that LLM reviewers exhibit conformity effects and limited diversity in reviewer–area-chair interactions. Our Study 1 extends this line from qualitative observation to paired statistical diagnosis: we quantify reject bias, minority elimination, flip asymmetry, and variance collapse under controlled conditions, and—critically—show that discussion provides no decision benefit while destroying diversity. Unlike AgentReview, which simulates the review process, our work uses the framework as both diagnostic instrument and treatment: AEIC's explicit Conclusion field makes herding measurable, and the Stackelberg layers supply the three loci of intervention—calibrated aggregation, behavioral regulation, and routing.

Structured Reasoning Decomposition. Chain-of-Thought prompting [33] elicits step-by-step reasoning, and Tree-of-Thoughts [35] extends this to structured exploration of reasoning paths. ReAct [36] interleaves reasoning with action. These methods decompose reasoning to improve *individual* answer quality. Our AEIC decomposition serves a different purpose: it structures agent outputs into four parseable fields (Assumption, Evidence, Inference, Conclusion) to enable *cross-agent* operations—reliable stance extraction from the explicit Conclusion field (+28pp over unstructured extraction), cross-topology voting-pattern comparison for adaptive routing, and population-level behavioral dynamics monitoring for the dynamic regulation mechanism (Section 4.6). The decomposition is not designed to improve per-agent reasoning; it is designed to make collective behavior observable and regulatable.

Communication Topology in Multi-Agent Systems. The effect of communication structure on multi-agent performance has been explored in several contexts. Chen et al. [5] use a centralized evaluator (star topology). Sequential architectures appear in chain-of-agents designs [14]. Debate topologies (complete graphs) are studied by Du et al. [9]. Chen et al. [4] introduce round-table conference among diverse LLMs with confidence-weighted voting. Most recently, RUMAD [30] applies reinforcement learning to adaptively control debate rounds and agent selection within a single topology type. Our approach differs in three ways: we select among five fundamentally different communication structures (parallel, sequential, star, debate, hierarchical), use interpretable voting-pattern features rather than opaque RL policies, and ground our mechanism in game-theoretic incentive compatibility guarantees.

Memory in Multi-Agent Systems. Several recent works examine how agent memory affects multi-agent dynamics. EquiMem [21] formulates shared memory in multi-agent debate as a zero-trust memory game where equilibrium calibrates optimal memory trust—a parallel game-theoretic approach to ours, but focused on memory integrity rather than consensus incentives. Liu et al. [20] demonstrate a paradoxical "memory curse" where expanded recall degrades cooperative intent in LLM-based social dilemmas (18 of 28 model–game configurations), showing that more historical information does not improve collective decision quality. This complements our finding (Section 7) that quality signals derived from AEIC structure cannot distinguish correct from misleading knowledge: both results indicate that richer per-agent information—whether historical memory or quality scores—does not necessarily improve consensus when agents share the same underlying LLM. Our dynamic regulation mechanism (Section 4.6) addresses this by operating on observable collective behavior rather than per-agent quality or memory content, making it robust to both signal limitations and memory quality variation.

Game Theory and LLMs. Sun et al. [27] provide a taxonomy of game theory applied to LLMs. Yi et al. [37] model multi-agent debate as a Bayesian game with Nash equilibria. Rauba et al. [24] argue for principal–agent modeling of

Table 1. Comparison with closest related work.

	Formal Incentives	Topology Selection	Sycophancy Solution	Structured Output	Behavioral Regulation
Standard Debate [9]	—	—	—	—	—
ConsensAgent [23]	—	—	Prompt-level	—	—
RUMAD [30]	—	✓†	—	—	—
Bertalanič et al. [2]	—	—	Analysis only	—	—
This work	✓‡	✓	✓	✓	✓

†Within-topology only (varying debate rounds/participation). ‡Stackelberg ϵ-incentive compatibility (Theorem 2).

LLM systems. Liu et al. [19] use LLMs for automated mechanism design. Most recently, Meng et al. [21] formulate shared memory in multi-agent debate as a zero-trust memory game where equilibrium calibrates optimal memory trust without assuming agent honesty, and Anand et al. [1] frame LLM agent agreement as a Byzantine consensus game, showing that prompted agents fail to reach agreement where classical theory guarantees convergence. Our Stackelberg formulation differs from both simultaneous-move equilibrium and Byzantine consensus approaches by modeling the mechanism designer as a leader who *commits* to aggregation rules before agents act—a natural fit for mechanism design where the goal is shaping follower behavior through credible commitment, independent of agent honesty assumptions or fault-tolerance guarantees.

Stackelberg Games in Distributed Systems. Stackelberg incentive mechanisms are established in federated learning [10,12] and crowdsourcing [32]. Saig et al. [25] apply contract theory to incentivize quality text generation from LLMs. Xu et al. [34] apply Stackelberg games to multi-agent contract revision. To the best of our knowledge, no prior work applies Stackelberg mechanism design to multi-agent LLM consensus with adaptive topology routing.

Positioning against Closest Work. Table 1 positions our framework against the four most closely related approaches.

3 AEIC Decomposition and Quality Signals

We introduce the AEIC decomposition, a structured reasoning protocol that enables reliable conclusion extraction across diverse communication topologies. The quality signals derived from AEIC components additionally serve as a formal interface for Stackelberg mechanism design (Section 4).

3.1 Structured Reasoning Decomposition

Given a question x, each agent $i \in \{1, \ldots, n\}$ produces a structured response decomposed into four components:

- **Assumption (A)**: The premise or hypothesis the agent adopts.
- **Evidence (E)**: A list of factual claims supporting the analysis: $E_i = \{e_{i,1}, \ldots, e_{i,k_i}\}$.
- **Inference (I)**: The reasoning chain connecting evidence to conclusion.
- **Conclusion (C)**: The final answer with justification.

This decomposition is enforced through prompt engineering and serves two purposes: (1) it structures agent output into parseable components, enabling reliable conclusion extraction and consistent stance identification across all communication topologies—this is the primary empirical value (Section 7); and (2) it provides separable components from which quality signals can be computed, serving as a formal interface for Stackelberg mechanism design (Section 4), though experiments reveal that these signals have limited discriminative power when agents share the same LLM backend (Section 7).

3.2 Four-Dimensional Quality Signals

From each agent's AEIC output, we compute four quality signals using embedding-based similarity:

Definition 1 (Quality Signals). *Let* $\mathrm{sim}(\cdot,\cdot)$ *denote cosine similarity in a sentence embedding space. For agent* i*:*

$$\mathrm{IC}_i = \mathrm{sim}(I_i, C_i) \tag{1}$$

$$\mathrm{EC}_i = \tfrac{1}{|E_i|} \textstyle\sum_{e \in E_i} \mathrm{sim}(e, C_i) \tag{2}$$

$$\mathrm{EQ}_i = f_{\mathrm{quality}}(E_i) \tag{3}$$

$$\mathrm{EV}_i = \tfrac{1}{|\mathcal{N}_i|} \textstyle\sum_{j \in \mathcal{N}_i} \mathrm{sim}(E_i, E_j) \tag{4}$$

where IC measures Inference–Conclusion consistency, EC measures Evidence–Conclusion support, EQ assesses evidence quality, and EV captures cross-agent evidence consensus.

Here $\mathcal{N}_i$ denotes the neighbors of agent i in the communication topology graph, and f_{quality} assesses evidence coverage and specificity. By design, IC, EC, and EQ measure *individual reasoning coherence*: how internally consistent, well-supported, and thorough an agent's analysis is. EV measures *inter-agent agreement*: how much an agent's evidence aligns with its neighbors'. This distinction matters for incentive design—EV can be inflated by sycophantic behavior. However, as our experiments demonstrate (Section 7), these coherence signals cannot distinguish correct from incorrect reasoning when agents share the same LLM: a coherent wrong answer scores as highly as a correct one.

3.3 Agent Type Space

Following Saig et al. [25], we model each agent through an observable *type*:

Definition 2 (Agent Type). *The type of agent i on question q is:*

$$\tau_i = (IC_i, EC_i, EQ_i) \in [0,1]^3 \tag{5}$$

EV is deliberately excluded from the type. It measures consensus rather than reasoning quality—a sycophantic agent can achieve high EV by mimicking neighbors without genuine reasoning. The type captures what the mechanism is *designed* to reward: individual reasoning effort. In practice, Section 7 shows that these type signals have limited discriminative power when all agents share the same LLM backend.

3.4 AEIC as Feature Extraction

Beyond mechanism design, AEIC-structured outputs provide the basis for topology routing features. For each question–topology pair, we construct a 42-dimensional feature vector from per-agent quality signals, summary statistics, *voting features* (majority ratio, yes/no distribution extracted from the Conclusion component), and cross-signal interactions. Feature ablation (Section 7) reveals that the voting features—enabled by reliable conclusion extraction from AEIC structure—are the primary driver of routing accuracy, while quality signals contribute marginally or negatively.

4 Stackelberg Consensus Game

4.1 Game Formulation

We formulate multi-agent consensus as a Stackelberg game $\mathcal{G} = (\mathcal{L}, \mathcal{F}, \Theta, \mathcal{S}, U_L, U_F)$.

Players. The *leader* $\mathcal{L}$ is the mechanism designer (bilevel optimizer). The *followers* $\mathcal{F} = \{1, \ldots, n\}$ are the LLM agents.

Strategy Spaces. The leader chooses mechanism parameters $\theta \in \Theta \subset \mathbb{R}^{21}$, controlling how AEIC quality signals map to consensus weights and how collective behavioral dynamics are regulated. Each follower i chooses a strategy $s_i \in \{s^{\text{ind}}, s^{\text{syc}}\}$:

- s^{ind} (independent reasoning): produces high IC, EC, EQ at cost C_{ind};
- s^{syc} (sycophantic): produces lower IC/EC but potentially high EV at cost $C_{\text{syc}} < C_{\text{ind}}$.

Timing.

1. Leader commits to parameters θ (mechanism rules).
2. Each follower observes θ and chooses strategy s_i.
3. Agents produce AEIC outputs; quality signals τ_i are computed.
4. Mechanism computes weights $w_i(\tau_i; \theta)$ via the scoring rule.
5. Weighted voting produces the consensus decision.

Utility Functions. The follower utility is:

$$U_i(s_i; \theta) = w_i(\tau_i(s_i); \theta) \cdot R - C(s_i) \tag{6}$$

where $R > 0$ is the reward for adoption. The leader maximizes:

$$\max_{\theta \in \Theta} \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{7}$$

4.2 Scoring Rule

The 21 parameters θ control three aspects:

- **Bid construction** (13 parameters): How IC, EC, EQ, EV signals combine into agent "bids" (alignment willingness α_i, revision cost, historical alignment h_i).
- **Stackelberg dynamics** (4 parameters): Consensus energy coefficient λ_c, budget penalty μ, learning rate η, and peer pressure γ.
- **Dynamic regulation** (4 parameters): Sycophancy detection threshold τ_h, deadlock threshold τ_l, EV decay rate α, and pressure adjustment rate β (Section 4.6).

Specifically, for agent i:

$$\alpha_i = \max\big(\theta_{\text{floor}},\ \text{IC}_i^{\theta_{\text{ic}}} \cdot \text{EQ}_i^{\theta_{\text{eq}}} \cdot (\theta_{\text{ev_base}} + \theta_{\text{ev_scale}} \cdot \text{EV}_i)\big) \tag{8}$$

$$h_i = \theta_{h_1} \cdot \text{IC}_i + \theta_{h_2} \cdot \text{EC}_i + \theta_{h_3} \cdot \text{EQ}_i + \theta_{h_4} \cdot \text{EV}_i \tag{9}$$

These bids are processed through a distributed consensus game that produces final weights $w_i(\theta)$ via iterative updates incorporating peer pressure from a graph Laplacian defined by the communication topology.

4.3 Incentive Compatibility Metric

We define incentive compatibility through inter-group weight comparison:

Definition 3 (Incentive Compatibility Metric). *For each question q, rank agents by quality score $\phi(\tau_i) = (IC_i + EC_i + EQ_i)/3$. Let H_q = top-2 agents and L_q = bottom-2 agents. Define:*

$$IC(\theta) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\frac{1}{|H_q|} \sum_{i \in H_q} w_i(\theta) > \frac{1}{|L_q|} \sum_{j \in L_q} w_j(\theta) \right] \tag{10}$$

This measures the fraction of questions where higher-quality reasoning receives greater consensus weight, capturing the essence of incentive compatibility.

4.4 Bilevel Optimization

The Stackelberg game is solved via bilevel optimization:

$$\textbf{Upper (Leader):} \quad \max_{\theta \in \Theta} \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{11}$$

$$\textbf{Lower (Followers):} \quad w_i = f(\tau_i; \theta) \quad \forall i \tag{12}$$

The lower-level problem is deterministic given θ (agents' AEIC outputs are fixed offline), reducing to single-level optimization implementable via differential evolution [26]. Per-topology θ_T^* is optimized with population size 21 ($1\times$ the number of parameters) and 30 generations, then evaluated via 5-fold stratified cross-validation to prevent train/test leakage. We note that the Stackelberg mechanism provides formal IC guarantees (Theorem 2), but experiments (Section 7) show that quality-weighted voting does not outperform majority voting under homogeneous LLM agents—the bottleneck is the quality signals' limited discriminative power, not the mechanism design.

4.5 Online Stackelberg Extension

In multi-round topologies (debate, hierarchical), the mechanism operates *online*: between rounds t and $t{+}1$, it computes interim AEIC signals from round-t outputs, runs the Stackelberg game to obtain current weights $w_i^{(t)}$, and injects quality feedback into agent prompts:

$$\text{context}_i^{(t+1)} = \text{context}_i^{(t+1)} \cup \{(\text{IC}_i^{(t)}, \text{EC}_i^{(t)}, w_i^{(t)})\} \tag{13}$$

This transforms the game from post-hoc reweighting to real-time behavioral shaping: agents observe their current quality assessment and can adjust reasoning effort accordingly. The star topology is inherently online—leaf trust scores computed via Stackelberg are injected into the hub's synthesis context.

Together with the static optimization of Section 4.4, this establishes a two-layer Stackelberg evolution: static optimization discovers *what* behaviors to suppress (e.g., near-zero EV weights on information-sharing topologies), while online feedback communicates *how* each agent is currently performing. However, the online extension injects the same quality feedback regardless of the collective behavioral pattern—it cannot distinguish sycophantic herding from productive convergence. Section 4.6 introduces a third layer that addresses this limitation.

4.6 Dynamic Sycophancy Regulation

Online Stackelberg feedback (Section 4.5) injects quality signals between rounds but applies a uniform intervention regardless of collective behavior. In practice, multi-round deliberation exhibits distinct pathological patterns: *sycophantic convergence*, where agents rapidly abandon independent positions to follow the emerging majority, and *deadlock*, where agents rigidly maintain initial positions without engaging with opposing evidence. We introduce a dynamic regulation layer that monitors collective stance dynamics and applies differentiated interventions.

Behavioral Metrics. Between rounds $t-1$ and t, we compute three metrics from agent conclusions $c_i^{(t)}$ (reliably extracted from the AEIC Conclusion field):

$$\mathrm{SCR}^{(t)} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\left[c_i^{(t)} \neq c_i^{(t-1)}\right] \tag{14}$$

$$\mathrm{MR}^{(t)} = \max\left(\frac{|\{i : c_i^{(t)} = \text{yes}\}|}{n},\ \frac{|\{i : c_i^{(t)} = \text{no}\}|}{n}\right) \tag{15}$$

$$\mathrm{CS}^{(t)} = \left|\mathrm{MR}^{(t)} - \mathrm{MR}^{(t-1)}\right| \tag{16}$$

where SCR (Stance Change Rate) measures the fraction of agents that changed their conclusion, MR (Majority Ratio) measures the dominance of the leading position, and CS (Convergence Speed) captures how rapidly the majority is growing.

State Classification. The system classifies each inter-round transition into one of three states:

- **Sycophantic** ($\mathrm{SCR} \geq \tau_h$): A large fraction of agents changed their stance, indicating potential herding behavior where agents abandon independent reasoning to follow the emerging majority.
- **Deadlock** ($\mathrm{SCR} \leq \tau_l \wedge \mathrm{MR} < 0.8$): Few agents changed stance and no clear majority exists, indicating entrenched positions without productive evidence exchange.
- **Healthy** (otherwise): Moderate stance evolution with either gradual convergence or stable disagreement—no intervention needed.

The thresholds τ_h, τ_l and the response parameters α, β (below) are included in the learnable parameter vector θ, extending the base 17 parameters to 21 dimensions. The bilevel optimization (Section 4.4) jointly discovers optimal detection thresholds and response intensities alongside the base Stackelberg parameters.

Adaptive Response. Upon detecting a non-healthy state, the mechanism responds through two channels:

1. **Parameter adjustment**: For sycophantic states, the evidence consensus weight is decayed: $\theta_{\text{ev_scale}} \leftarrow \theta_{\text{ev_scale}} \cdot (1-\alpha)$, suppressing the very signal that rewards conformity. Peer pressure and consensus energy are similarly reduced: $\gamma \leftarrow \gamma \cdot (1-\beta)$, $\lambda_c \leftarrow \lambda_c \cdot (1-\beta)$. For deadlock, these parameters are *increased* to encourage engagement. Both $\alpha \in [0.1, 0.9]$ and $\beta \in [0.1, 0.6]$ are optimized by the leader.
2. **Differentiated prompts**: The next-round context is augmented with state-specific instructions: anti-sycophancy prompts emphasize independent reasoning over majority-following; exchange prompts encourage engaging with opposing evidence; neutral prompts provide standard deliberation guidance.

This completes a three-layer Stackelberg evolution: *static* optimization discovers which behaviors to suppress (Section 4.4), *online* feedback communicates individual quality assessments (Section 4.5), and *dynamic* regulation detects and corrects collective behavioral pathologies in real time. The key insight is that the Stackelberg game's value lies in *collective behavior regulation*—shaping how the group deliberates—rather than in quality-based selection of individual contributions.

5 Adaptive Topology Routing

A key insight of this work is that no single communication topology dominates across all questions. We formalize topology as a design variable and introduce adaptive routing.

5.1 Communication Topologies

We define five communication topologies as weighted adjacency matrices $W \in \mathbb{R}^{n \times n}$:

- **Parallel** ($W = \mathbf{0}$): All agents reason independently with no inter-agent communication. This is the standard self-consistency setting.
- **Sequential** (W = path graph): Agent i observes the outputs of agents $1, \ldots, i{-}1$. Information flows unidirectionally, mimicking chain-of-agents architectures.
- **Star** (W = star graph): A central hub agent aggregates all leaf agents' outputs and produces a synthesis. Leaf agents reason independently.
- **Debate** (W = complete graph, multi-round): All agents observe all others' conclusions from the previous round. Repeated for 3 rounds, simulating multi-agent debate.
- **Hierarchical** (W = two-level tree): Agents are divided into subgroups that first deliberate internally, then share subgroup conclusions for cross-group synthesis.

The topology W affects both signal computation (EV and agreement signals depend on $\mathcal{N}_i$) and agent behavior (context-aware generation in sequential, star, debate, and hierarchical modes).

5.2 Per-Topology Parameter Optimization

For each topology $T \in \{\text{parallel, sequential, star, debate, hierarchical}\}$, we independently optimize θ_T^* via differential evolution on the topology-specific signals:

$$\theta_T^* = \arg\max_{\theta \in \Theta} \ \text{Accuracy}_T(\theta) + \lambda \cdot \text{IC}_T(\theta) \tag{17}$$

This yields five specialized consensus configurations. The optimized parameters reveal topology-specific structure (Figure 1): star and debate topologies

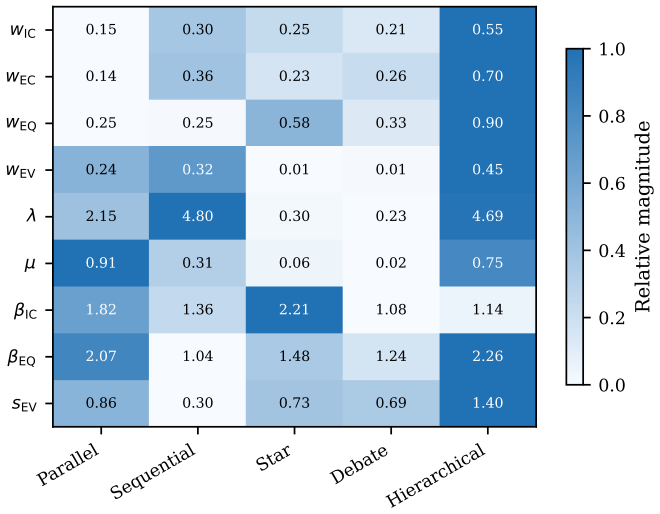


Fig. 1. Optimized per-topology parameters θ_T^* (QA-domain validation). Star and debate learn near-zero EV weights—an emergent anti-sycophancy configuration; hierarchical and sequential learn high consensus pressure λ_c.

learn near-zero weights for EV ($w_{\text{EV}} \approx 0.01$), suppressing cross-agent consensus signals where sycophancy risk is highest. Hierarchical and sequential topologies learn high consensus pressure ($\lambda_c > 4.5$), enforcing agreement within their chain/tree structures.

5.3 Neural Topology Router

To select the optimal topology per question, we train a neural router using features derived from AEIC-structured outputs.

Feature Design. For each candidate topology T, we construct a 42-dimensional feature vector comprising per-agent quality signals (IC, EC, EQ, EV, agreement for 5 agents), summary statistics (mean, std, range), *voting features* (majority ratio, yes/no distribution extracted from AEIC conclusions), and cross-signal interactions (*e.g.*, $\text{IC}_i \times (1-\text{agreement}_i)$). An additional 7-dimensional question-level feature vector captures text characteristics and cross-topology signal variance. As Section 7 shows, the voting features—enabled by AEIC's structured conclusion extraction—are the primary driver of routing accuracy (+2.1pp when present), while several quality signal groups contribute marginally or negatively.

Router Architecture. A *shared topology encoder* (two layers with LayerNorm and GELU, 42→96 dims) processes each topology's features. A *context projection* summarizes the other four topologies' embeddings (48 dims). Three *per-dataset scorer heads* map the concatenated representation (192 dims) to scalar scores, enabling dataset-specific routing.

Training. We use a focal ranking loss with adaptive margins:

$$\mathcal{L} = \sum_{(c,w)} \left(\frac{v}{m}\right)^{\gamma} \cdot v, \quad v = \max(0,\ m - (s_c - s_w)) \tag{18}$$

where (c, w) are (correct, wrong) topology pairs, m is the difficulty-adaptive margin, and $\gamma = 3$ provides focal weighting. Training uses AdamW with cosine annealing over 300 epochs, 5-fold stratified cross-validation.

Top-1 Selection. The router selects the highest-scoring topology per question as the consensus topology. With per-dataset specialized scoring heads, this achieves 87.6% ± 0.3% accuracy across 5 seeds (5-fold cross-validation), compared to 83.1% for the best fixed topology (star). Correctness labels for training use majority vote outcomes rather than Stackelberg-weighted decisions, avoiding contamination from optimized θ^* parameters.

6 Theoretical Analysis

Theorem 1 (Existence of Stackelberg Equilibrium). *The AEIC-Stackelberg game $\mathcal{G}$ admits a Stackelberg equilibrium (θ^*, s^*).*

Proof. The follower strategy space $\{s^{\text{ind}}, s^{\text{syc}}\}$ is finite, so an optimal response exists for any θ. The parameter space Θ is compact (bounded). The leader's objective $\text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta)$ is bounded on Θ. By the Weierstrass extreme value theorem, the maximum is attained at some $\theta^* \in \Theta$. Together with the followers' best responses, this constitutes a Stackelberg equilibrium.

Theorem 2 (ϵ-Incentive Compatibility). *For any $\epsilon \in (0, 1)$, setting $\lambda \geq \frac{1}{1-\epsilon}$ in Eq. (7) guarantees IC$(\theta^*) \geq \epsilon$.*

Proof. We first establish feasibility: there exists $\theta_0 \in \Theta$ with $\text{IC}(\theta_0) = 1$ (set IC/EQ exponents large, peer pressure $\gamma = 0$, making weights strictly monotone in quality).

By optimality of θ^*:

$$\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_0) + \lambda \cdot 1 \tag{19}$$

Rearranging with $\text{Acc}(\theta_0) - \text{Acc}(\theta^*) \geq -1$:

$$\text{IC}(\theta^*) \geq 1 - \frac{1}{\lambda} \tag{20}$$

Setting $\lambda \geq \frac{1}{1-\epsilon}$ yields $\text{IC}(\theta^*) \geq \epsilon$.

Theorem 3 (Individual Rationality and Welfare). *(a) Under θ^*, all agents achieve non-negative utility when $R \geq C_{ind}/\min_i w_i$.*
(b) Accuracy$(\theta^*) \geq$ Accuracy$(\theta_{uniform})$ for sufficiently small λ.

Proof. (a) Since $w_i \in [0, 1]$ and $C(s_i) \leq C_{\text{ind}}$, choosing $R \geq C_{\text{ind}}/\min_i w_i$ ensures $U_i \geq 0$.

(b) Uniform voting corresponds to $\theta_{\text{unif}} \in \Theta$. By optimality: $\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}}) + \lambda \cdot \text{IC}(\theta_{\text{unif}})$. For $\lambda \to 0$, this directly yields $\text{Acc}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}})$.

Convergence of Distributed Consensus. Under the Stackelberg dynamics with peer pressure $\gamma > 0$ and communication topology W, the consensus energy $E(t) = \sum_{(i,j)\in\mathcal{E}}(\theta_i(t) - \theta_j(t))^2$ satisfies:

$$E(t) \leq E(0) \cdot \exp\left(-\frac{2\gamma t}{n-1}\right) \tag{21}$$

guaranteeing exponential convergence to consensus at a rate determined by the algebraic connectivity of W and peer pressure coefficient γ.

7 Experiments

We validate the framework on automated scholarly review through a diagnose-then-intervene design: Study 1 establishes the pathologies of LLM review panels (Section 7.2) and their cross-model generality (Section 7.3); Study 2 applies the framework's three layers as targeted interventions (Sections 7.4–7.6).

7.1 Experimental Setup

Dataset. We assemble 300 papers from the ICLR 2024 conference on OpenReview [22], stratified into three tiers of 100 by official decision and mean reviewer score: *clear-accept* (Oral/Spotlight), *borderline* (poster with mean score nearest to the acceptance boundary), and *clear-reject* (lowest-scoring rejections). Each paper contributes its title, abstract, keywords, official decision, and human reviewer scores. The official decision serves as a noisy ground truth; we therefore treat decision agreement as a *safety* endpoint and behavioral health as the *primary* endpoint (below).

AEIC-Review protocol. Each reviewer agent produces a structured review instantiating the AEIC decomposition (Section 3): Assumption (the paper's claimed contributions), Evidence (a list of (+)/(−)-tagged strengths and weaknesses anchored to the paper), Inference (overall reasoning), and Conclusion (a stance prefix `Accept`/`Weak Accept`/`Weak Reject`/`Reject` plus a 1–10 rating). The explicit Conclusion field makes stance and rating extraction reliable across models; reviewer personas differ only in decoding temperature. This field is the measurement basis for every behavioral metric in this paper: stances and ratings are read off structured fields, never inferred from free text.

Conditions. Each paper is reviewed by a homogeneous panel of 5 reviewer agents under three conditions: **parallel** (independent reviews, single round), **debate** (three rounds; after round 1 each reviewer observes anonymized peer positions and may revise), and **star** (independent reviews plus a meta-reviewer synthesis). Study 2 adds **debate-regulated** (Section 7.5). Default panels use DeepSeek-V3 [7]; cross-model panels use GLM-4-Flash [11] and DeepSeek-R1 [8] on a matched 150-paper subset (first 50 per stratum).

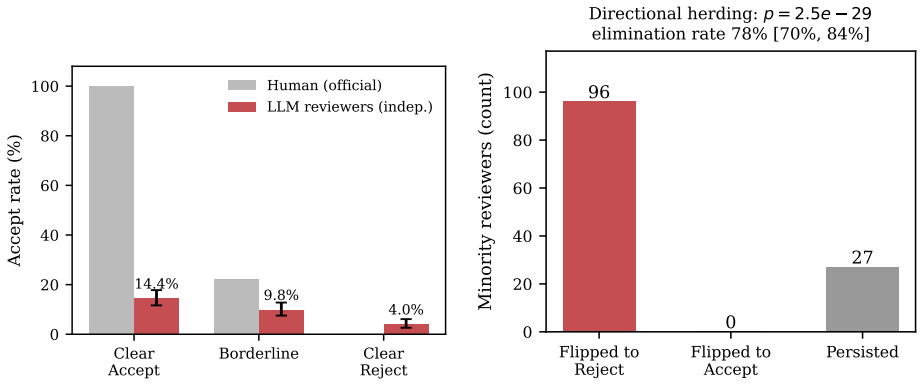


Fig. 2. Study 1 diagnosis. Left: independent accept rate per stratum vs. the official base rate—a systematic reject bias. Right: fate of round-1 minority positions after three discussion rounds: flips run 96:0 toward rejection.

Metrics. Behavioral health (primary): minority elimination rate and flip direction (an r1 minority reviewer is *eliminated* if their final stance equals the r1 majority; otherwise they *persist*); rating variance collapse $(1-\bar{\sigma}_{r3}/\bar{\sigma}_{r1})$; mean rating drift. *Decision quality* (safety): agreement of panel decisions with the official outcome, overall and per stratum. *Calibration*: MAE of panel mean rating vs. human mean score. Statistics: exact binomial test (flip direction), Wilcoxon signed-rank (paired continuous), McNemar (paired decisions), Wilson 95% CIs.

7.2 Study 1: Diagnosis of LLM Review Panels

Independent reviews carry a systematic reject bias. On clear-accept papers (officially Oral/Spotlight), independent DeepSeek-V3 reviewers accept only 14.4% of the time; the borderline and clear-reject strata sit at 9.9% and 4.0% (Figure 2). The bias is a property of the single-agent prior, not of aggregation—and it is exactly the pathology that Study 2a's calibrated layer targets.

Discussion triggers a negative ratchet. Across the 90 panels with a non-unanimous first round, minority positions are eliminated at 78.0% (Wilson 95% CI [69.9, 84.5]), and flips are entirely one-directional: 96 toward rejection, 0 toward acceptance, 27 persist (binomial p=2.5×10^{-29}). Panel rating variance collapses by 51.1% (Wilcoxon p=3.9×10^{-32}) and mean ratings drift by −0.25 points (p=1.0×10^{-32}). Roughly 70% of panels are already unanimous after round 1, so discussion predominantly compresses—it rarely deliberates.

Discussion yields no decision benefit. Decision agreement with the official outcome is statistically flat across conditions: parallel 61.7%, debate r1 61.3%, debate r3 59.7%, star meta-review 60.7% (all pairwise McNemar p>0.05). Three rounds of discussion thus destroy diversity without buying decision quality—the

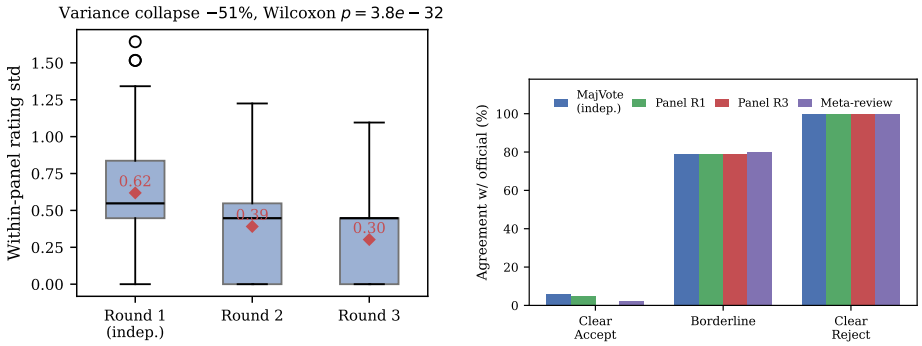


Fig. 3. Left: per-paper rating std, round 1 vs. round 3—variance collapses by 51%. Right: decision agreement by condition—discussion and meta-review add nothing over independent aggregation.

signature pathology that Study 2c's routing layer exploits. Figure 3 summarizes the collapse and the flat decision profile.

7.3 Cross-Model Generality: Opposite Pathological Attractors

We replicate the debate condition on a matched 150-paper subset with homogeneous panels of GLM-4-Flash and DeepSeek-R1 (Figure 4). Herding is universal—every model eliminates ≥83% of minorities—but its *direction* is model-dependent. DeepSeek-V3 and R1 panels ratchet toward rejection (100% and 90% of flips; accept baserates ≤19%), whereas GLM-4-Flash panels ratchet toward *acceptance* (0% of flips toward rejection; accept baserate ≥97.6%; 94.4% of conclusions written as hedged "Accept/Reject" forms whose stance must be derived from the rating). Most strikingly, the strongest reasoner does not herd least: R1 achieves the best decision agreement (67.3%) yet the largest variance collapse (76.4%), versus 54.0% for V3 and 12.3% for GLM (whose panels begin near-unanimous). Capability does not protect against herding; it merely selects the attractor. GLM's near-total conformity also caps its decision quality at 38.7%, 28.6pp below R1.

7.4 Study 2a: Calibrated Aggregation Repairs the Reject Bias

The static Stackelberg layer optimizes aggregation parameters θ on a stratified train split (150 papers): per-reviewer weights w_i, signal weights on evidence specificity and (+)/(−) balance, a calibration bias τ, and a decision threshold, maximizing agreement minus $\lambda\times$calibration MAE (differential evolution; test-only reporting on the held-out 150). Learned θ^* exhibits a strong positive calibration $\tau = +1.04$—the optimizer's direct answer to the diagnosed reject bias—and upweights evidence-specific reviews ($\gamma_1 = +1.09$).

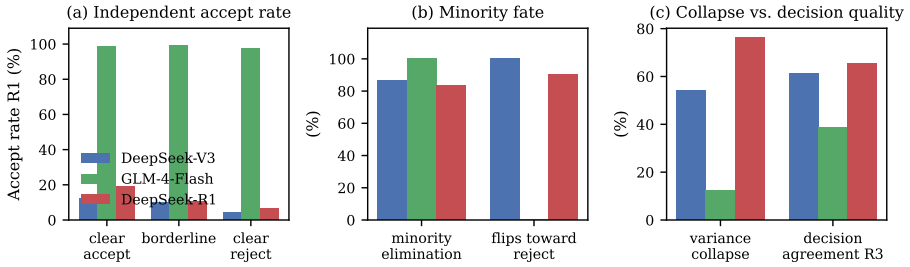


Fig. 4. Cross-model replication on the matched 150-paper subset. (a) Independent accept rate per stratum: V3 and R1 are reject-biased, GLM is accept-biased. (b) Minority elimination is universal (≥83%), but flip direction is model-dependent. (c) The strongest reasoner (R1) suffers the largest variance collapse.

Table 2. Test-set decision agreement with official outcomes (n=150). θ^* repairs the clear-accept stratum from near-zero; most of the gain is already captured by re-thresholding the plain mean (trained threshold). MAE vs. human mean score.

Method	Overall	Clear-acc.	Borderline	Clear-rej.	MAE
Majority vote	60.0%	2.0%	78.0%	100%	—
Mean threshold	66.7%	44.0%	70.0%	86.0%	1.18
Meta-review (star)	60.0%	—	—	—	—
θ^* aggregation	64.7%	38.0%	68.0%	88.0%	1.14

Table 2 shows the result. Against majority vote, θ^* lifts clear-accept agreement from 2.0% to 38.0% and overall agreement by +4.7pp (McNemar b=20, c=13, p=0.30). A honest baseline check tempers the claim: re-thresholding the plain mean reaches 66.7% overall. In a *homogeneous* pool, calibration—in any form—is necessary and nearly sufficient; reviewer- and signal-level weighting adds little, consistent with the QA-domain finding that quality signals carry little discriminative power when agents share one backend.

7.5 Study 2b: Prompt-Level Regulation Cannot Break the Ratchet

The dynamic layer instantiates direction-neutral, evidence-anchored regulation: whenever a panel is non-unanimous, minority reviewers are instructed to change position only if peers rebut specific Evidence items (and to cite the rebutting evidence if they do), while majority reviewers must ground persuasion in specific rebuttals rather than headcounts. On the 89 of 300 panels where the intervention fires (173 minority injections), elimination drops only 78.0%→71.0%, variance collapse is unchanged (51.1%→52.4%), and intervened minorities persist at 29% versus 28% for matched unregulated controls; flips still run 87:1 toward rejection. Decision agreement is preserved (59.7%→60.7%)—the safety endpoint holds. On GLM panels the intervention fires rarely (7% of panels, since round-1 unanimity

Table 3. Test-set routing vs. fixed protocols (n=150). The router matches the best fixed protocol at less than half the deliberation cost; the oracle shows headroom that round-1 features do not unlock.

Policy	Agreement	Avg. calls/paper
Fixed θ^* independent	64.7%	5.0
Fixed debate	58.7%	15.0
Fixed debate-regulated	60.7%	15.0
Fixed meta-review	60.0%	6.0
Rule baseline (unanimous→independent)	60.0%	6.0
Router (argmax)	64.7%	6.6
Router (cost-sensitive)	64.7%	5.3
Oracle	74.7%	5.3

is near-total) and never rescues a minority (0/7). We report this as a negative result with a clear implication: *rhetorical* anchoring cannot overcome capability-driven conformity; effective regulation must act on the aggregation mechanism or the deliberation structure, not on phrasing.

7.6 Study 2c: Routing Demotes Discussion to an On-Demand Resource

Since ∼70% of panels are unanimous after round 1, deliberation is wasted on most papers. The routing layer selects per paper among four protocols—θ^* independent aggregation (5 calls), unregulated debate (15), regulated debate (15), meta-review (6)—using round-1-prospective features (stance entropy, rating mean/std, evidence specificity, evidence balance, inference length) with per-action logistic scorers estimating P(correct) on the train split.

Table 3 shows that the router ties the best fixed protocol (64.7%) at 6.6 calls per paper—a 56% cost reduction versus always discussing (13.4× fewer calls than the 15-call protocols on 83% of papers)—and its action mix (124/150 θ^*-independent, 22 regulated-debate, 4 others) directly reflects the Study 1 finding that discussion rarely adds decision value. The 10pp oracle gap indicates headroom that round-1 features cannot yet unlock, a direction for richer prospective signals. Figure 5 summarizes all three interventions.

8 Discussion

Herding as a pathological attractor, not a bug of phrasing. Study 1 and the cross-model replication converge on a structural picture: multi-round LLM panels do not merely converge—they converge toward a model-specific attractor. For DeepSeek-V3 and R1 the attractor is rejection; for GLM-4-Flash it is acceptance with hedged language. The attractor's pull is strong enough that an explicit, direction-neutral moderation protocol (Study 2b) changes almost nothing: minorities instructed to persist unless specifically rebutted still capitulate

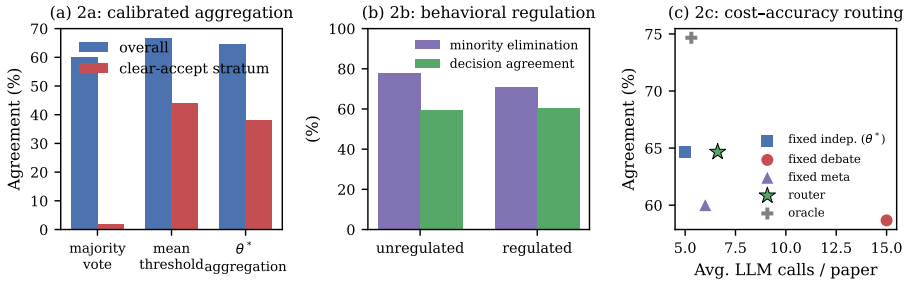


Fig. 5. Study 2 interventions. (a) Calibrated aggregation repairs the clear-accept stratum. (b) Prompt-level behavioral regulation barely moves minority elimination while preserving decision agreement. (c) Routing matches the best fixed protocol at the cost frontier.

at the same rate as unmoderated controls. This mirrors the QA-domain finding that static anti-sycophancy must act on mechanism *parameters* (EV-weight suppression) rather than prompts, and extends it: even behavioral prompting fails when the underlying model prior is conformist. Effective regulation of LLM panels must therefore operate at the aggregation layer or the deliberation-structure layer, not the rhetoric layer.

A calibration-first doctrine. The clearest positive result is also the cheapest: the reject bias—a single-agent prior distortion—is repaired at aggregation time. The Stackelberg-optimized θ^* discovers this on its own, placing a +1.04 calibration term on the consensus score, and the simpler re-thresholded mean captures nearly the same agreement. The practical doctrine: before adding deliberation, calibrate the aggregation. Deliberation without calibration amplifies the prior's bias through the ratchet; calibrated independent aggregation already beats every discussion protocol we tested on decision agreement per call.

Discussion as an on-demand resource. Across two domains now, the same pattern appears: most instances need no discussion (70% of panels are unanimous after round 1; on QA benchmarks, most questions are topology-insensitive), and unguided discussion can actively hurt. Routing converts this observation into cost savings: the review-domain router spends deliberation budget on only 17% of papers and matches the best fixed protocol. The 10pp oracle gap marks the frontier—better prospective features (paper difficulty signals, reviewer disagreement semantics beyond entropy) could unlock it.

Behavioral health as the primary endpoint. A methodological contribution of this work is the endpoint doctrine. Decision agreement barely moved across any condition or intervention (all within 59–65%), yet behavioral metrics moved by 50+ points with $p<10^{-28}$. If we had measured only decisions, Study 1 would have reported "discussion is harmless"—exactly the wrong conclusion. Any evaluation of multi-agent judgment systems should report minority survival, variance

preservation, and drift alongside accuracy, with accuracy demoted to a safety constraint. This is especially important when ground truth is noisy, as with official review decisions.

Transfer beyond peer review. The three layers map onto general multi-LLM judgment pipelines. Calibrated aggregation applies wherever multiple signals fuse into a decision—ranking ensembles in recommendation, multi-critic evaluation, red-team voting—wherever a shared prior bias needs a learned correction term. The behavioral health metrics (elimination rate, variance collapse, drift, flip asymmetry) constitute a *deliberation health monitor* for multi-agent pipelines: they are computable from logged stances alone and could serve as production observability signals for agentic systems that deliberate before acting. Protocol routing generalizes to compute allocation: which queries merit multi-model deliberation and which should be answered by a single calibrated call. We have validated none of these transfers empirically; they are the natural next studies.

Relation to the QA-domain validation. Collective judgment poses two questions—*is the decision right* and *is the process healthy*—and the framework's two validations answer one each. On QA benchmarks, the same framework validated the routing layer (87.6% accuracy, +4.5pp over the best fixed topology) and the structured-decomposition premise of AEIC (+28pp from explicit Conclusion fields), while its quality signals proved to measure coherence rather than correctness. The review domain changes the question: with noisy ground truth and behavioral stakes, the framework's value shifts from accuracy gains to diagnosis (making herding measurable), calibration (repairing prior bias), and cost-aware orchestration. The two validations are complementary: QA shows the framework optimizes what can be optimized; review shows it regulates what must be protected.

Limitations. (1) Panels are homogeneous per model; heterogeneous panels (mixed models, mixed personas) may herd differently—our three-model study samples backends but does not mix them within a panel. (2) The official decision is noisy ground truth; we mitigate via the safety-endpoint doctrine and per-stratum reporting, but absolute agreement values should be read as "agreement with human panels," not correctness. (3) GLM stances are 94% rating-derived (hedged conclusions), so its behavioral metrics partly reflect rating dynamics. (4) Study 2b tests one specific prompt-level protocol; stronger mechanism-level regulations (e.g., aggregation-time minority protection, structured rebuttal rounds) remain untested. (5) The router's features are round-1 statistics; prospective difficulty estimation is shallow, and the oracle gap suggests substantial headroom. (6) All datasets derive from one venue and year (ICLR 2024); cross-venue generality is open.

9 Conclusion

We subjected a game-theoretic multi-agent consensus framework—AEIC decomposition, a three-layer Stackelberg mechanism, and adaptive protocol routing—to a diagnose-then-intervene validation on automated scholarly review. The diagnosis is unambiguous: LLM review panels herd. Independent reviews carry a systematic reject bias (14.4% accept rate on clear-accept papers); unstructured discussion triggers a negative ratchet that eliminates 78% of minority positions with flips running 96:0 toward rejection (p=2.5×10^{-29}), collapses rating variance by 51% (p=3.9×10^{-32}), and never improves decision agreement. Herding is universal across three model families, but its direction is model-dependent—DeepSeek-V3 and R1 ratchet toward rejection while GLM-4-Flash ratchets toward acceptance—and the strongest reasoner suffers the largest diversity collapse: capability does not protect against herding, it selects the attractor.

The interventions map mechanism class to pathology. Aggregation-time calibration, discovered autonomously by Stackelberg optimization as a +1.04 bias term, repairs the reject bias (clear-accept agreement 2%→38%)—and a re-thresholded mean matches it, so *calibrate first, in whatever form.* Prompt-level behavioral regulation cannot break the ratchet (elimination 78%→71%) though it preserves decision quality: rhetoric is no match for a conformist prior, and effective regulation must act on mechanism or structure. Per-paper routing matches the best fixed protocol at 56% lower cost, demoting deliberation to an on-demand resource. Throughout, behavioral health metrics proved orders of magnitude more sensitive than decision accuracy; we advocate minority survival, variance preservation, and drift as primary endpoints for any multi-agent judgment system, with accuracy demoted to a safety constraint.

Future work includes mechanism-level regulation that protects minorities at aggregation time rather than in prompts, heterogeneous panels mixing model families (whose opposite attractors may cancel), richer prospective features to close the 10pp oracle gap in routing, and transfer of the deliberation health metrics to production multi-agent pipelines—recommendation ensembles, multi-critic evaluation, and agentic systems that deliberate before acting—as observability signals.

Disclosure of Interests. The authors have no competing interests to declare.

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